

Unit 4, Lesson 3: Sequences of Transformations



Benchmarks

MA.912.GR.2.2 Identify transformations that do or do not preserve distance.

MA.912.GR.2.3 Specify a sequence of transformations that will map a given figure onto itself, or onto another congruent or similar figure.



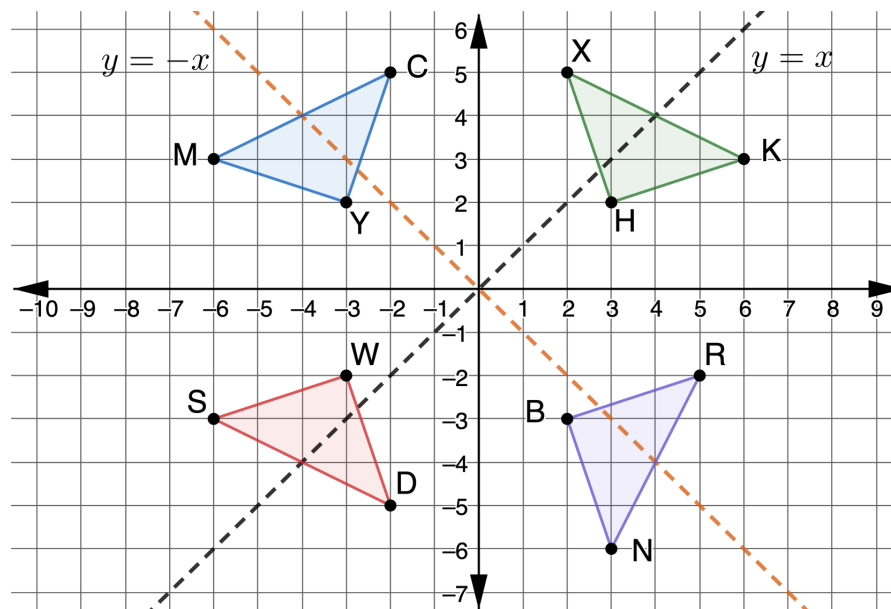
Learning Targets

- ☐ I can identify transformations that preserve distance.
- ☐ I can specify a sequence of transformation that will map a given figure onto another congruent figure.



4.3.1 Warm-Up: Bell Work

Triangles MCY , KXH , NRB , and SWD are shown on the coordinate grid.



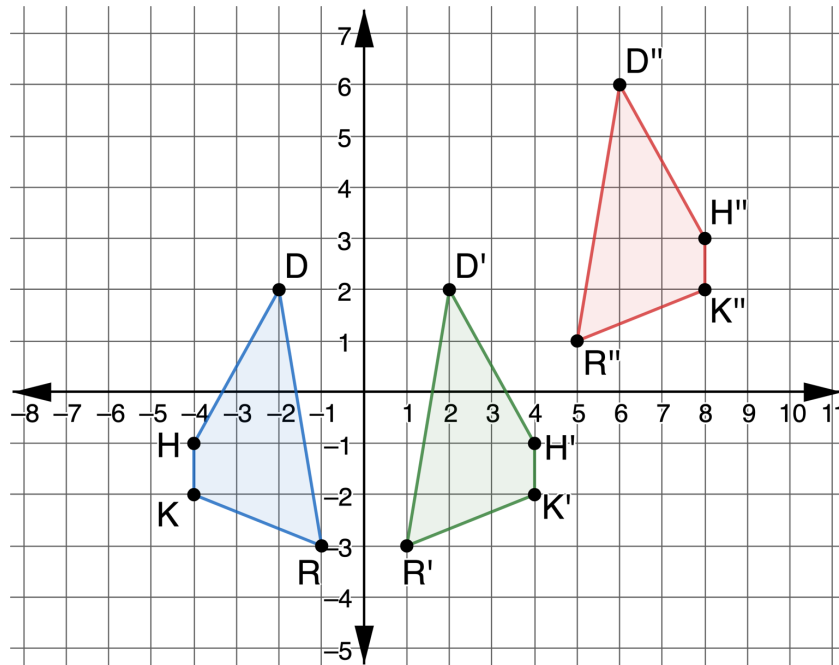
Complete each statement.

- When $\triangle MCY$ is reflected across _____, it will map onto $\triangle KXH$.
- When $\triangle NRB$ is reflected across _____, it will map onto $\triangle$ _____.
- When $\triangle SWD$ is reflected across the x -axis, it will map onto $\triangle$ _____.



4.3.2 Guided Instruction: Sequence of Transformations

1. Quadrilateral $RKHD$ was transformed to create quadrilateral $R'K'H'D'$. Then, quadrilateral $R'K'H'D'$ was transformed to create quadrilateral $R''K''H''D''$. The quadrilaterals are shown on the coordinate grid.



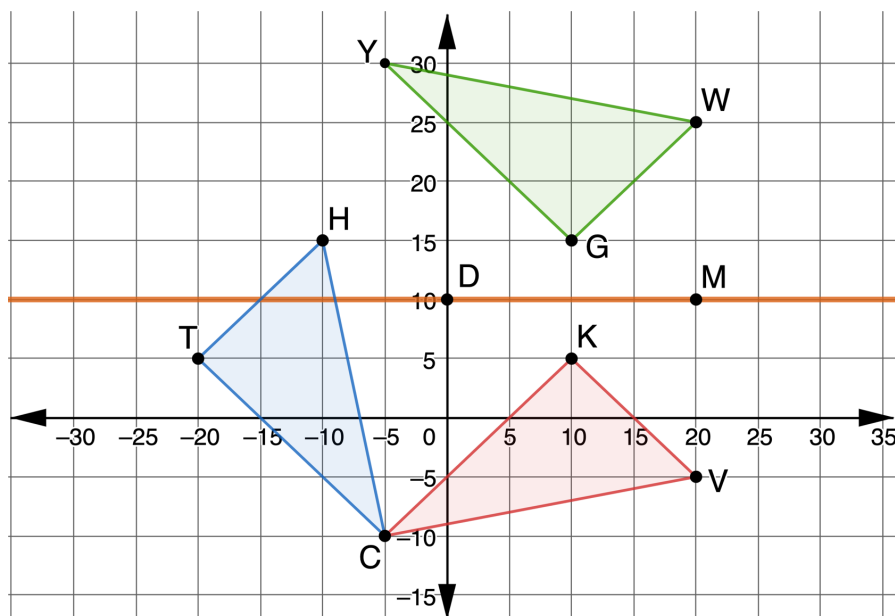
- a. On the graph, show the mappings of the vertices of the first transformation.
- b. On the graph, show the mappings of the vertices of the second transformation.

- c. Complete the table.

	Quadrilateral $RKHD$ Transformed to Quadrilateral $R'K'H'D'$	Quadrilateral $R'K'H'D'$ Transformed to Quadrilateral $R''K''H''D''$
Written Description		
Algebraic Description		

- d. Explain how to know if a sequence of transformations preserves distance.

2. Triangles HTC , VKC , and WGY are shown on the coordinate grid. Triangle HTC is the pre-image. Triangle VKC is the result of the first transformation, and $\triangle WGY$ is the result of the second transformation.



a. Describe the first transformation.

b. Use patty paper to show that the transformation preserved distance.

c. Describe the second transformation.

d. Use patty paper to show that the second transformation preserved distance.

e. Work with your partner to complete the statement.

Because the ☐ center of rotation ☐ line of reflection of ☐ $\triangle HTC$ ☐ $\triangle VKC$

is ☐ $(0, 0)$, ☐ $(-5, -10)$, ☐ $y = 0$, ☐ $y = 10$, an algebraic description

☐ can ☐ cannot be used.

f. When Judeline and Carmen completed the statement above, they had different correct answers. Write the second correct answer.

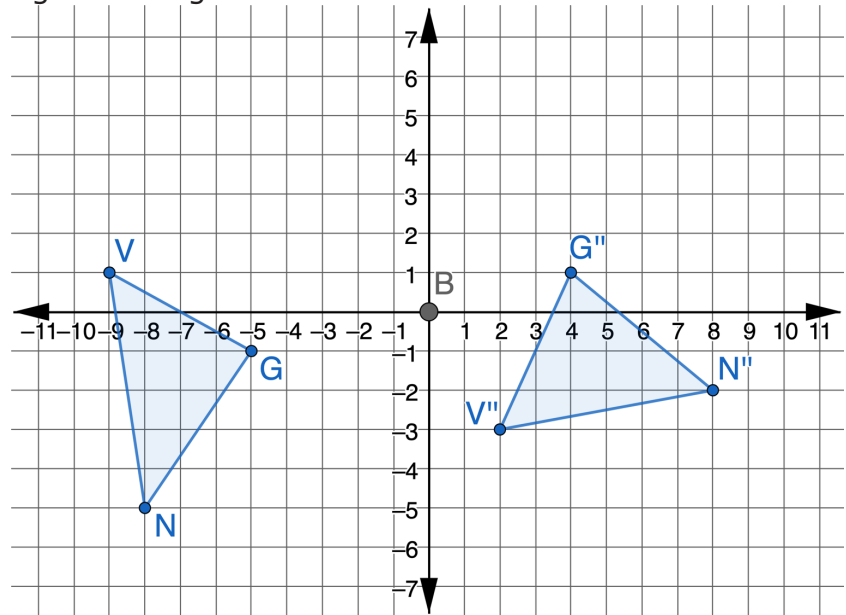


To use an algebraic description of a rotation the center of rotation must be _____.



To use an algebraic description of a reflection the line of reflection must be _____, _____, or _____.

3. Triangles NVG and $N''V''G''$ are shown on the coordinate grid. Triangle NVG was transformed to create $\Delta N''V''G''$.



- a. Explain which features of the two triangles can be used to determine that the transformations were rigid motions.
- b. Adele told Kiriam that she knows one of the rigid motions is a rotation. Discuss with your partner which clues in the location of the vertices Adele used to draw her conclusion.

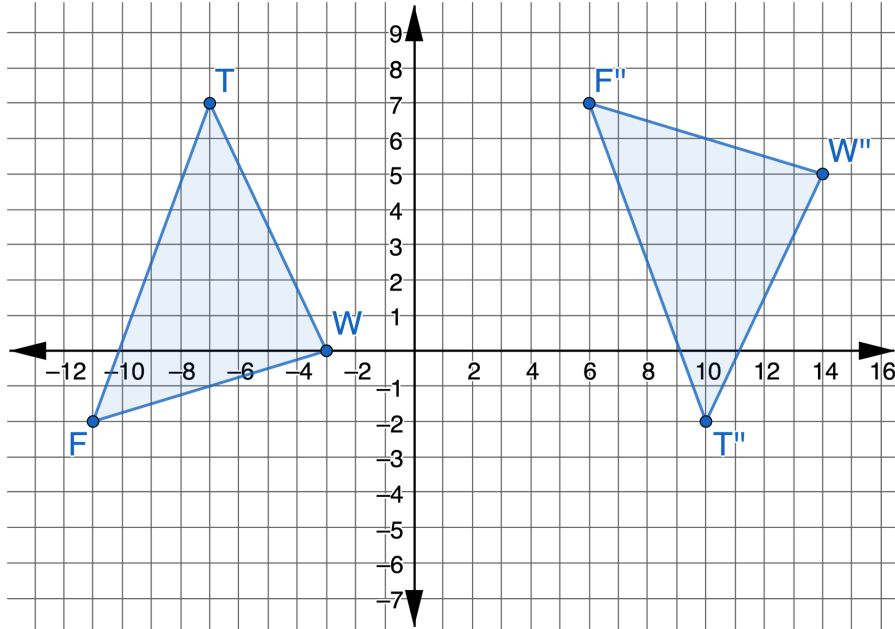
- c. Work with your partner to determine a sequence of transformations that includes a rotation about the point B so that when applied, ΔNVG will map onto $\Delta N''V''G''$. Complete the table.

Transformation	Mapping	Algebraic Description
	$\Delta NVG \rightarrow \Delta N'V'G'$	
	$\Delta N'V'G' \rightarrow \Delta N''V''G''$	



4.3.3 Guided Practice: Sequence of Transformations

- Triangles TWF and $T''W''F''$ are shown on the coordinate plane.



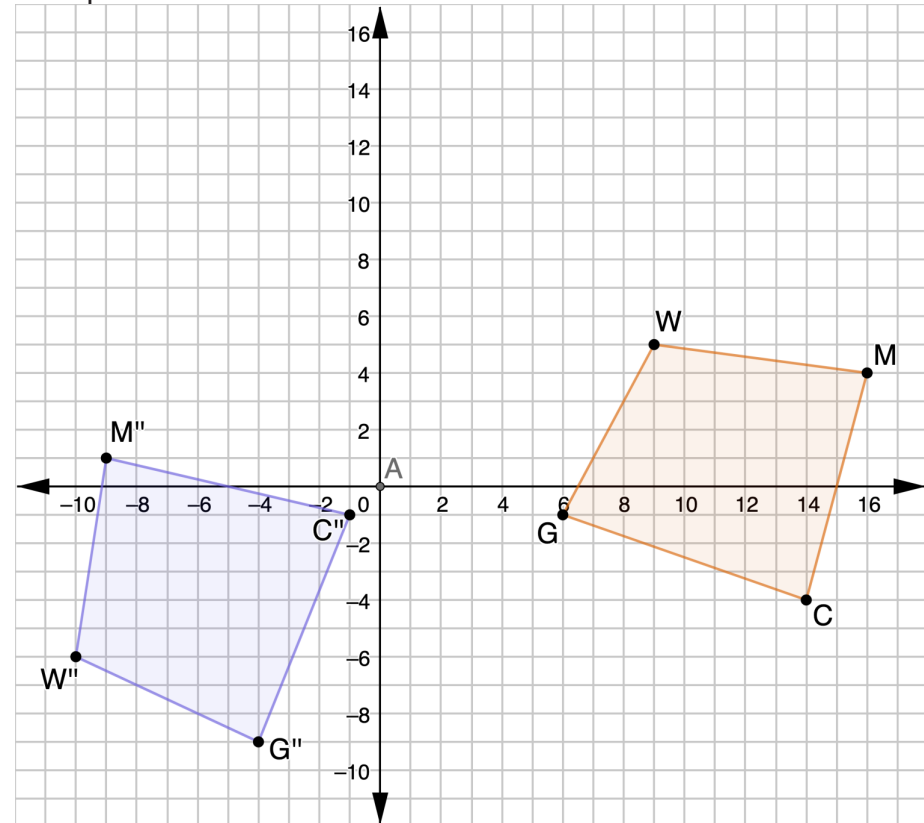
Determine a sequence of transformations that will map $\triangle TWF$ onto $\triangle T''W''F''$. Complete the table.

Transformation	Mapping	Algebraic Description
	$\triangle TWF \rightarrow \triangle T'W'F'$	
	$\triangle T'W'F' \rightarrow \triangle T''W''F''$	



4.3.4 Your Turn!

- Quadrilateral $MCGW$ was transformed to create quadrilateral $M''C''G''W''$.



- Describe the sequence of transformations that will map quadrilateral $MCGW$ onto quadrilateral $M''C''G''W''$.
- Write the sequence of transformations using algebraic descriptions.



4.3.5 Wrap-Up: Reflection

1. Check the box that describes your understanding of each learning target.

- ☐ I can identify transformations that preserve distance and I am ready for the next lesson.

- ☐ I need some help identifying transformations that preserve distance. One area in which I need help is

_____.

I need more support in knowing how to identify transformations that preserve distance. I am

- ☐ concerned I will not be ready to move on. One area in which I need help is

_____.

- ☐ I can specify a sequence of transformation that will map a given figure onto another congruent figure.

I need some help specifying a sequence of transformation that will map a given figure onto another congruent figure. One area in which I need help is

_____.

- ☐ I need more support in knowing how to identify transformations that preserve distance. I am concerned I will not be ready to move on. One area in which I need help is

_____.



Unit 4, Lesson 4: Written Descriptions of a Sequence of Transformations



Benchmarks

MA.912.GR.2.1 Given a preimage and image, describe the transformation and represent the transformation algebraically using coordinates.

MA.912.GR.2.5 Given a geometric figure and a sequence of transformations, draw the transformed figure on a coordinate plane.



Learning Targets

- ☐ I can draw an image given a pre-image and a written description of a sequence of transformations on a coordinate plane.
- ☐ I can describe a transformation on the coordinate plane using written descriptions.